Time: 1 hr. B. Tech. (5th Semester) 1st Sessional Test 2025-26 Max. Marks: 12

Paper Code: CSE-501 Introduction to Data Mining

Note: Attempt All Question

Q1. CO1 Describe in detail the architecture of data mining. Also discuss the different types of data on which data mining can be applied. [4] [BL-1]

Q2. CO1 Given the following data (in ascending order) for the attribute age: [4] [BL-2,3]

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52,70.

a) Use smoothing by bin means to smooth these data, using equal frequency bins of size 3. Illustrate the steps and comment on the effect of this technique for the given data.
b) How might you determine outliers in the data? What other methods are there for data smoothing?
c) Use min-max normalization to transform the value 35 for age onto the range [0.0,1.0]
d) Use Z-score normalization to transform the value 35 for age.

Q3. CO2 A database has five transactions. Let minimum_support = 60% and minimum_confidence = 80%. [4] [BL-3]

| TID | Items Bought |
|---|---|
| T100 | M, O, N, K, E, Y |
| T200 | D, O, N, K, E, Y |
| T300 | M, A, K, E |
| T400 | M, U, C, K, Y |
| T500 | C, O, O, K, I, E |

Find all the frequent items using Apriori algorithm. Show all the steps of algorithm. How we can improve the performance of Apriori algorithm. Discuss?